MILITARY SPECIFICATION

SPIRONOLACTONE AND HYDROCHLOROTHIAZIDE TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Spironolactone and Hydrochlorothiazide Tablets (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3)

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.

3. REQUIREMENTS

3.1 Material. Shall be Spironolactone and Hydrochlorothiazide Tablets and shall be accordance with all applicable requirements of FED-STD-140, and as specified herein.

S2. Classification. Shall be type I.

S5.3 The following is added to this paragraph:

Shall be tablets containing 25 mg of Spironolactone and 25 mg of Hydrochlorothiazide, within the designated assay limits of the tablets.

S6.4.1 Tablets may be mottled.

S6.4.9.1 Shall meet the requirements of the USP Disintegration Test for Uncoated Tablets in not more than 20 minutes.

S6.4.10 In addition to compling with the USP Weight Variation Test, tablets shall comply with the USP content uniformity requirements based on individual assay of tablets for spironolactone content.

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3.1.1 Assay. The tablets shall contain not less than 93.0 percent and not more than 107.0 percent of the required amounts of Spironolactone and Hydrochlorothiazide when assayed as specified in 4.4.1.

3.2 Ingredients.

3.2.1 Spironolactone and hydrochlorothiazide. The spironolactone and hydrochlorothiazide used in the manufacture of the tablets shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.2.2 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.3 Expiration dating period. The expiration dating period shall be 60 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling as shown in FED-STD-140 shall be applicable.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Labeling inspection. Sampling shall be in accordance with MIL STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.5 Tests.

4.5.1 Assay for spironolactone and hydrochlorothiazide.

Weigh and finely powder 20 tablets. Accurately weigh a portion of the powder equivalent to about one (1) tablet and transfer to a 200 ml volumetric flask. Add about 100 ml of methanol to the flask, and, while swirling, heat just to boiling on a steam bath. Cool to room temperature and make to volume with methanol. Mix thoroughly and then filter through a No. 2 Whatman filter paper discarding the first 15 ml of the filtrate. Pipet 5 ml of the clear filtrate into a 50 ml volumetric flask and make to volume with methanol. Mix well. Dissolve an accurately weighed quantity of USP Spironolactone Reference Standard in methanol and quantitatively and stepwise dilute with methanol to obtain a standard solution having a concentration of about 12.5 mcg per ml. Dissolve an accurately weighed quantity of USP Hydrochlorothiazide Reference Standard in methanol and quantitatively and stepwise dilute with methanol to obtain a standard solution having a concentration of about 12.5 mcg per ml.

Concomitantly determine the maximum absorbances of the sample and the Spironolactone standard solution in 1 cm cells, at the wavelength of approximately 238 nm with a suitable spectrophotometer, using Hydrochlorothiazide standard solution as a blank.

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Calculate the percent of the required amount as follows:

$$\frac{Au}{As} \times C \times \frac{50}{5} \times \frac{200}{25} \times \frac{\text{Ave Tablet Wt.}}{\text{Sample Wt.}} \times 100 = \text{percent of Spironolactone}$$

Where: Au is the absorbance of the sample
As is the absorbance of the Spironolactone standard solution
C is the concentration of the Spironolactone standard solution in mg per ml.

Concomitantly determine the maximum absorbance of the sample and Hydrochlorothiazide standard solution in 1 cm cell at the wavelength of approximately 269 nm with a suitable spectrophotometer using Spironolactone reference standard as a blank.

Calculate the percent of the required amount as follows:

$$\frac{Au}{As} \times C \times \frac{50}{5} \times \frac{200}{25} \times \frac{\text{Ave Tablet Wt.}}{\text{Sample Wt.}} \times 100 = \text{percent of Hydrochlorothrazide}$$

Where: Au is the absorbance of the sample
As is the absorbance of the Hydrochlorothiazide standard solution
C is the concentration of the Hydrochlorothiazide standard solution in mg per ml

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3), and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing 100 tablets, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

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5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1)

5.3.1 Level A. Two hundred eighty-eight (288) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Two hundred eighty-eight (288) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

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NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend. "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock Number	Item Identification
6505-00-782-1107	SPIRONOLACTONE AND HYDROCHLOROTHIAZIDE TABLETS, 100s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1655

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OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers, vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced documents(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M-38748 MODIFIED SHEET, USP

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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